CubeSat UD34 – PCB P2 OBC-ADCS

Esquemático completo (Práctica 4) – proyecto Altium P2_OBC-ADCS.PrjPcb

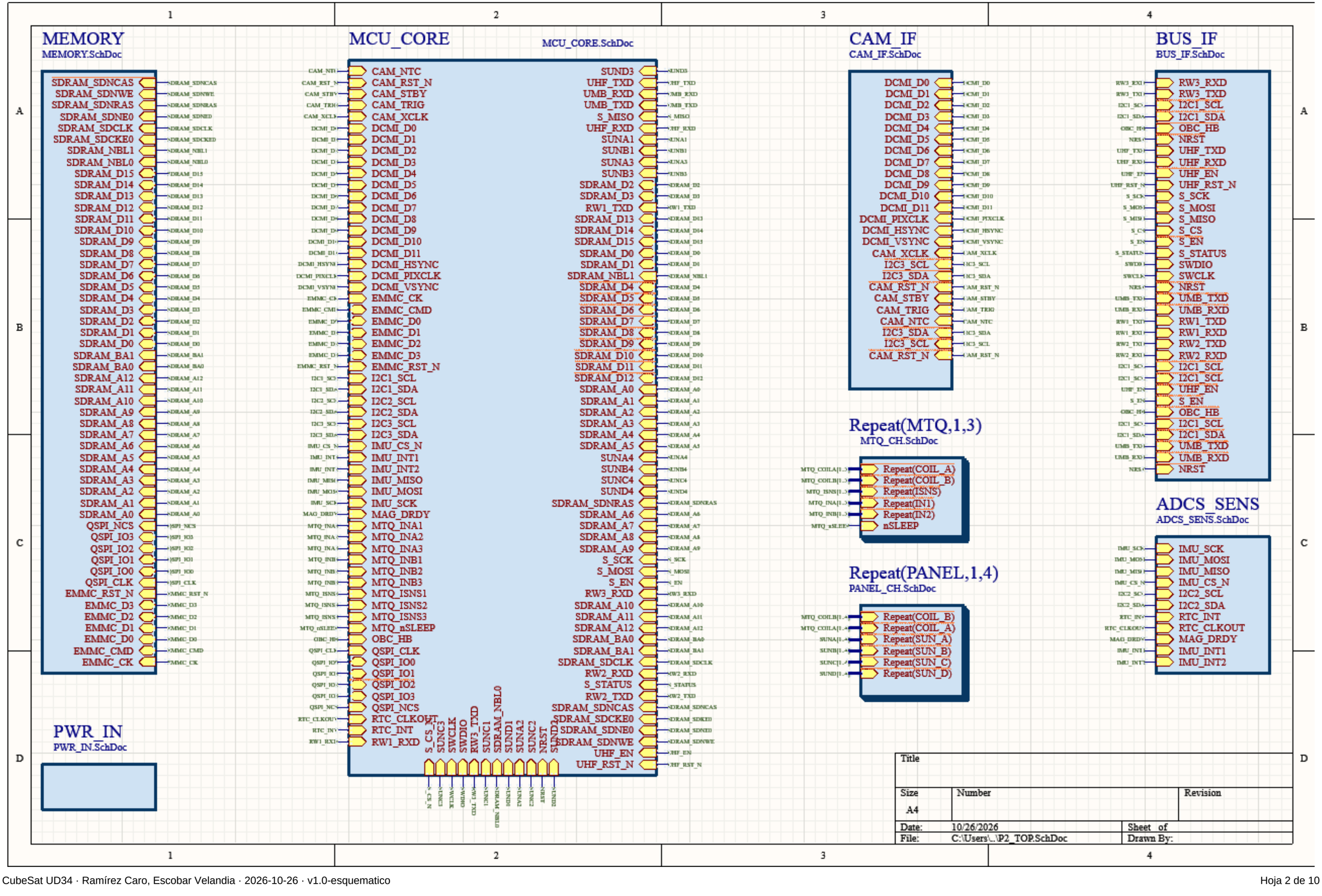
Autores: Duvan Camilo Ramírez Caro · Daniela Escobar Velandia

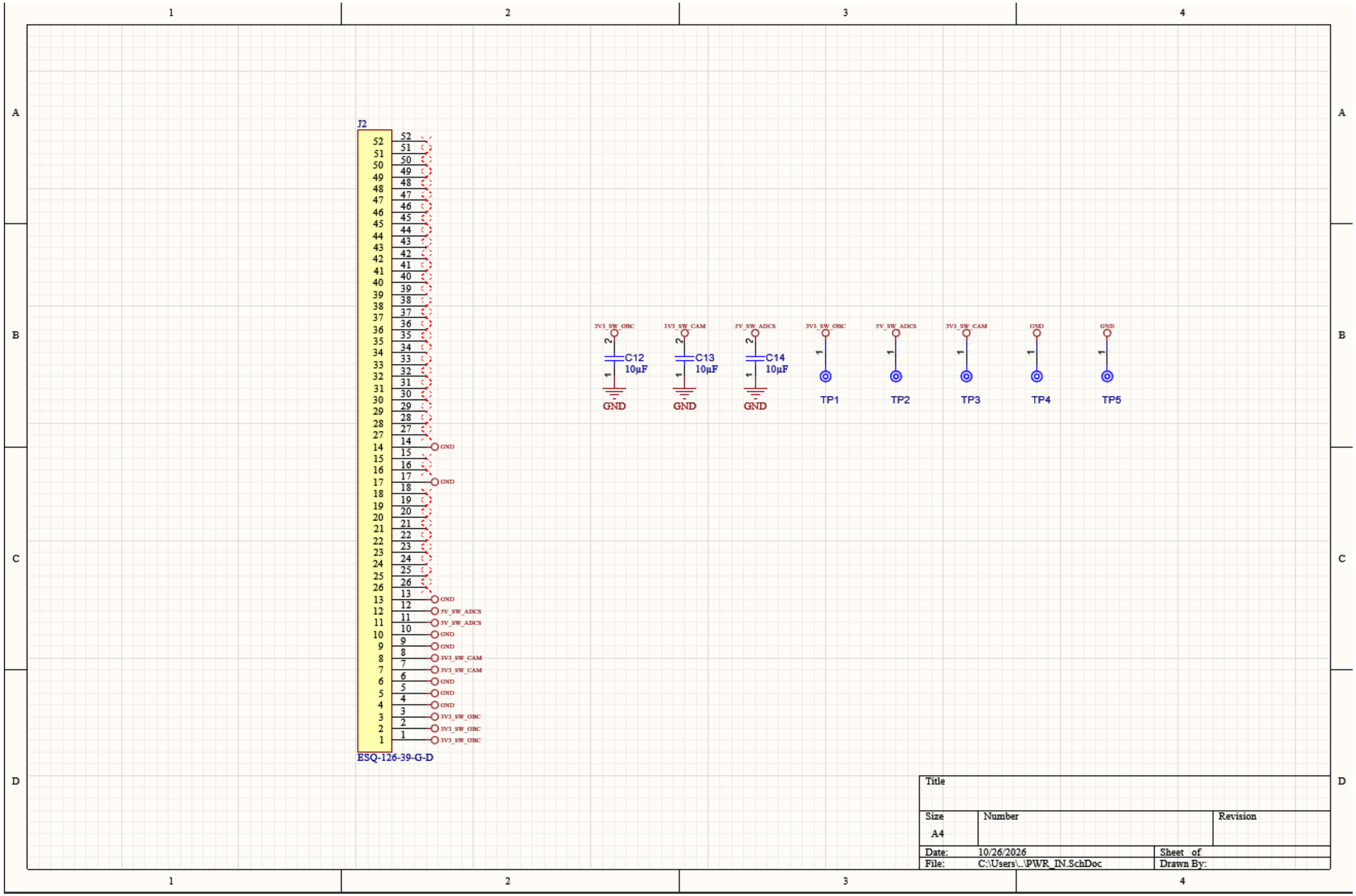
Fecha: 2026-10-26 · Revisión: v1.0-esquematico

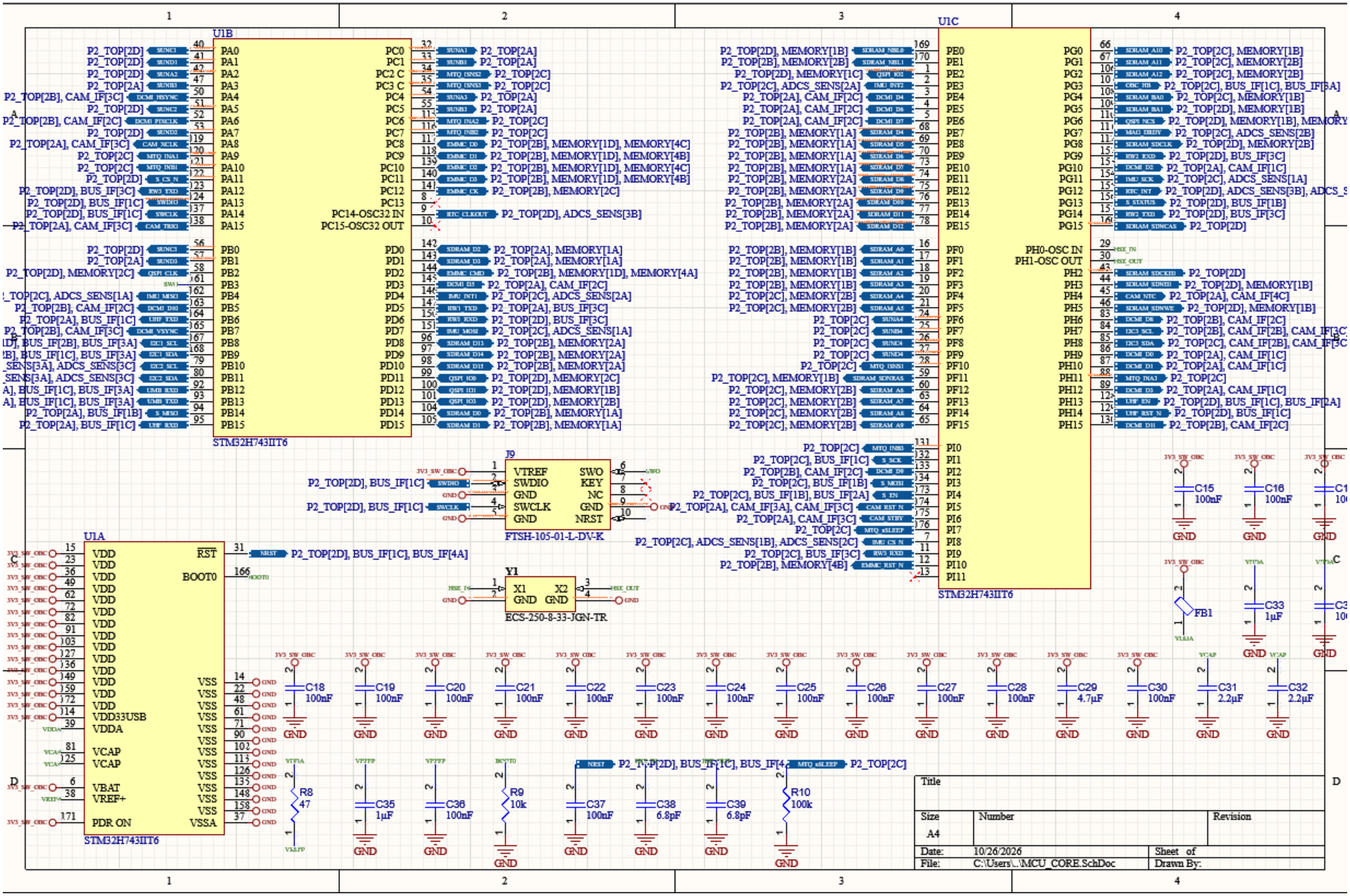
Contenido

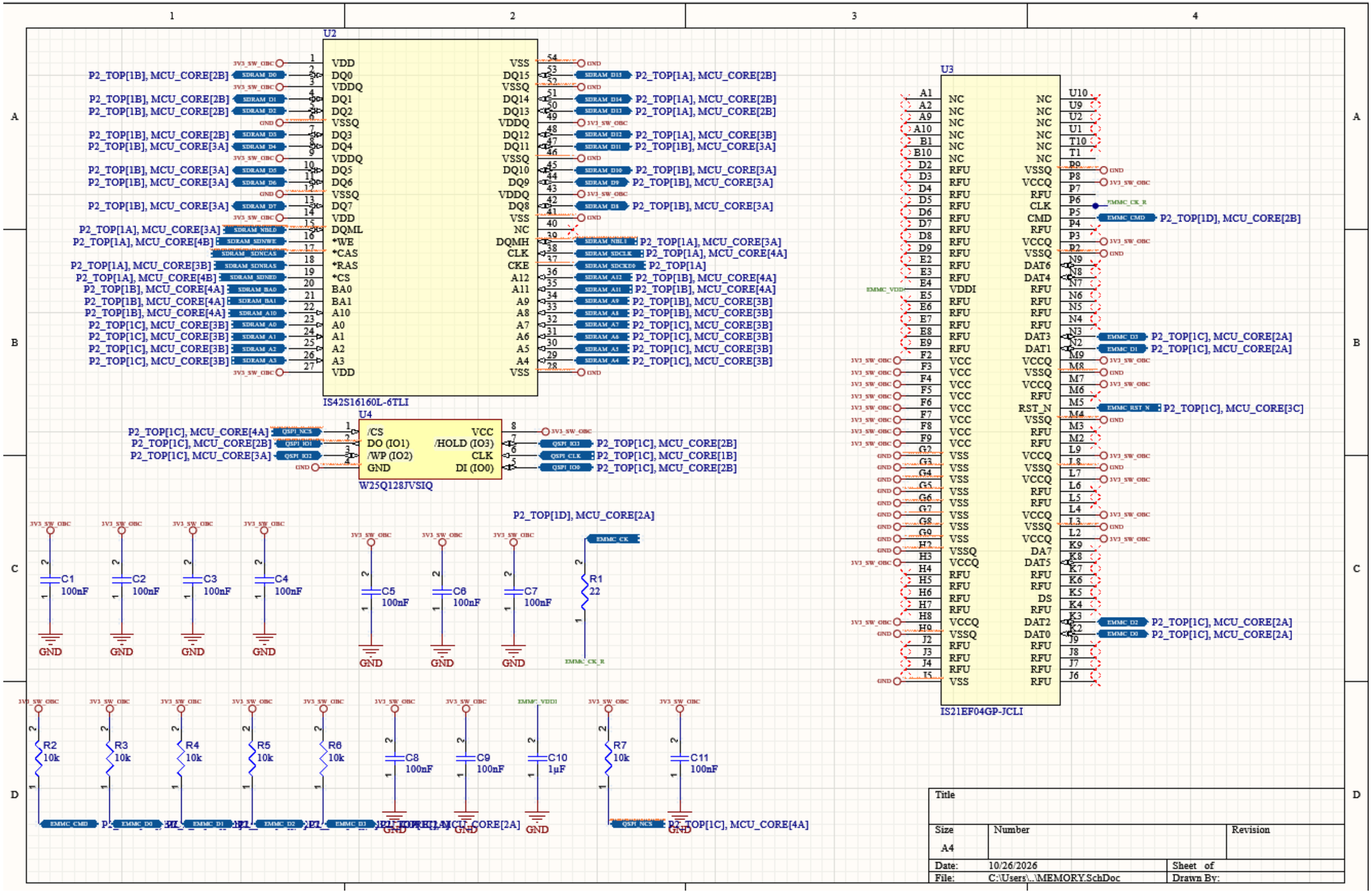
Hoja 2/10 P2_TOP.SchDoc – Hoja principal: bloques e interconexiones (multicanal Repeat(MTQ,1,3) y Repeat(PANEL,1,4))
Hoja 3/10 PWR_IN.SchDoc – Entrada de potencia: J2 (H2 del PC/104), desacoplo de rieles y puntos de prueba
Hoja 4/10 MCU_CORE.SchDoc – MCU STM32H743IIT6 (U1A, U1B, U1C), cristal HSE, SWD (J9), desacoplo, VDDA/VREF+
Hoja 5/10 MEMORY.SchDoc – SDRAM IS42S16160L (FMC), eMMC IS21EF04GP (SDMMC1) y NOR W25Q128JV (QUADSPI)
Hoja 6/10 CAM_IF.SchDoc – Interfaz de cámara: J3 FFC de 40 pines a la P4, pull-ups I2C3
Hoja 7/10 ADCS_SENS.SchDoc – Sensores: IMU ISM330DHCX (SPI1), magnetómetro LIS2MDL y RTC RV-3032 (I2C2)
Hoja 8/10 BUS_IF.SchDoc – Bus del satélite J1 (H1 del PC/104), ruedas de reacción J8, pull-ups y puntos de prueba
Hoja 9/10 MTQ_CH.SchDoc – Canal de magnetorquer (×3): DRV8837 + INA180A2 + shunt 0,47 Ω
Hoja 10/10 PANEL_CH.SchDoc – Canal de panel (×4): conector JST GH, filtros RC de sensores solares, bobina

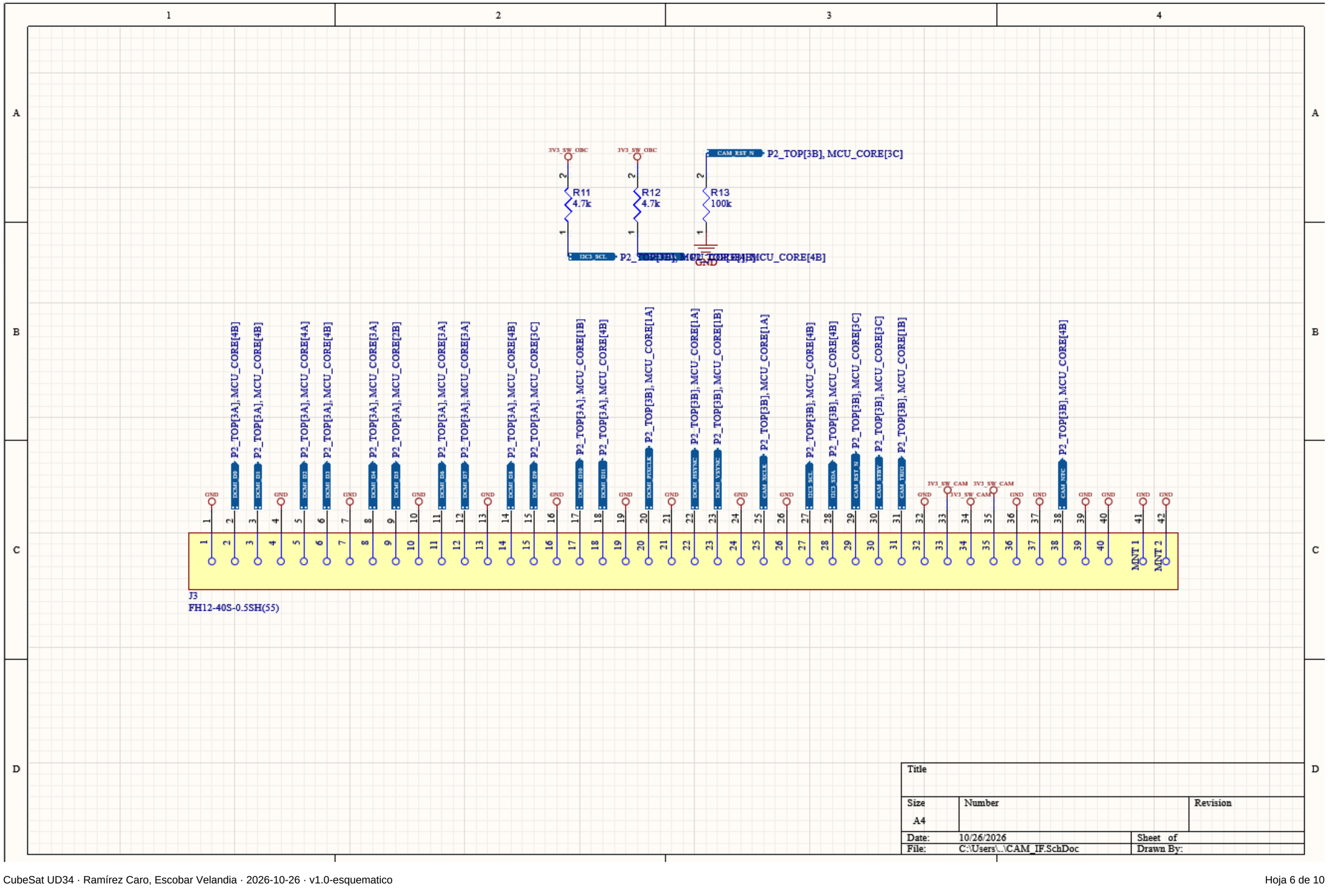
Nota: las hojas de este documento son capturas de pantalla del editor de Altium Designer (2026-10-26), no la exportación Smart PDF.
Los canales repetidos (MTQ1–MTQ3, PANEL1–PANEL4) se muestran con su hoja única; la versión exportada se reemplazará en la siguiente revisión.

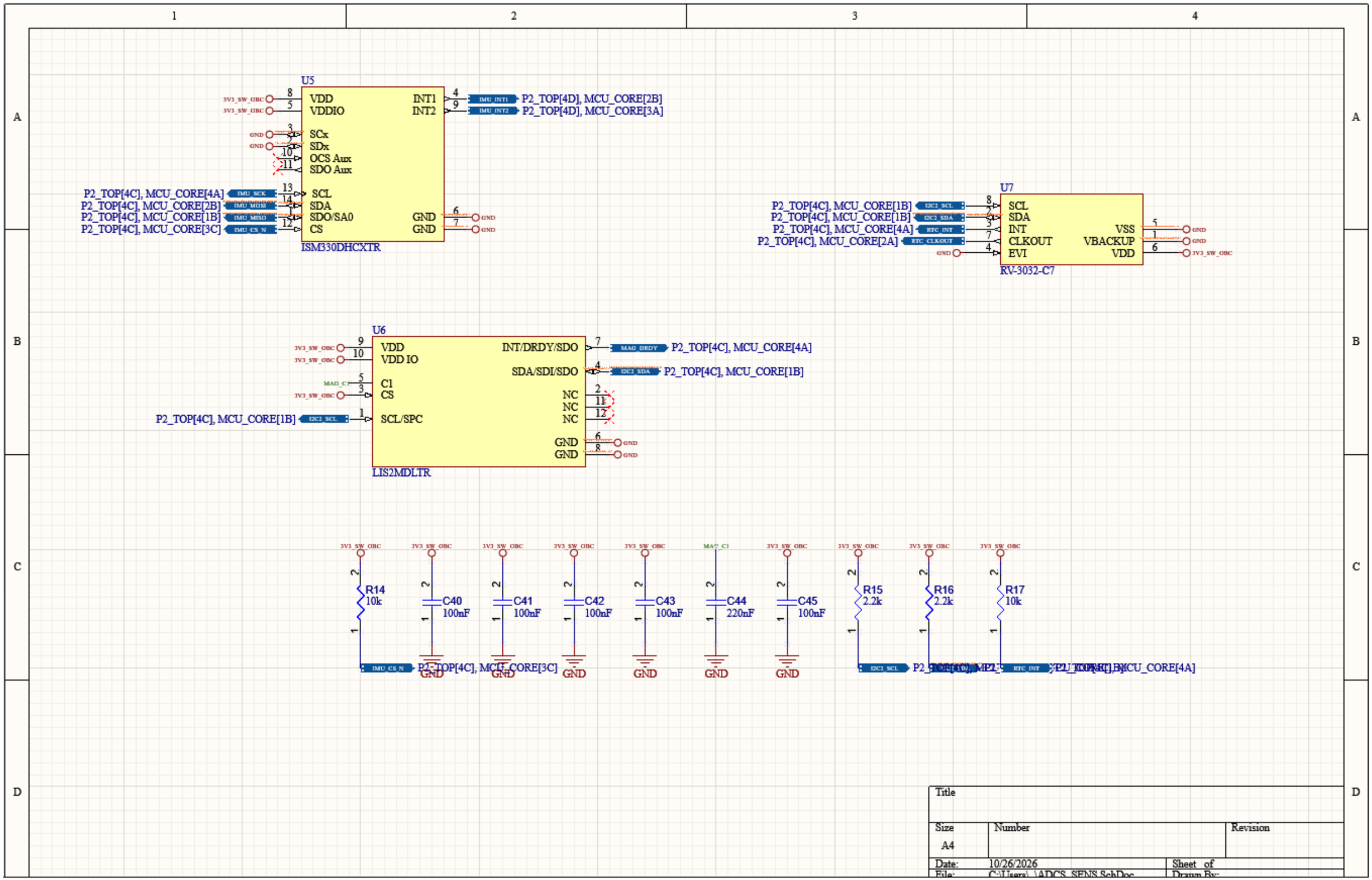

MEMORY
MEMORY.SchDoc
SDRAM_SDNCAS
SDRAM_SDNWE
SDRAM_SDNRAS
SDRAM_SDNE0
SDRAM_SDCLK
SDRAM_SDCKE0
SDRAM_NBL1
SDRAM_NBL0
SDRAM_D15
SDRAM_D14
SDRAM_D13
SDRAM_D12
SDRAM_D11
SDRAM_D10
SDRAM_D9
SDRAM_D8
SDRAM_D7
SDRAM_D6
SDRAM_D5
SDRAM_D4
SDRAM_D3
SDRAM_D2
SDRAM_D1
SDRAM_D0
SDRAM_BA1
SDRAM_BA0
SDRAM_A12
SDRAM_A11
SDRAM_A10
SDRAM_A9
SDRAM_A8
SDRAM_A7
SDRAM_A6
SDRAM_A5
SDRAM_A4
SDRAM_A3
SDRAM_A2
SDRAM_A1
SDRAM_A0
QSPI_NCS
QSPI_IO3
QSPI_IO2
QSPI_IO1
QSPI_IO0
QSPI_CLK
EMMC_RST_N
EMMC_D3
EMMC_D2
EMMC_D1
EMMC_D0
EMMC_CMD
EMMC_CK
MCU_CORE
MCU_CORE.SchDoc
CAM_NTC
CAM_RST_N
CAM_STBY
CAM_TRIG
CAM_XCLK
DCMI_D0
DCMI_D1
DCMI_D2
DCMI_D3
DCMI_D4
DCMI_D5
DCMI_D6
DCMI_D7
DCMI_D8
DCMI_D9
DCMI_D10
DCMI_D11
DCMI_HSYNC
DCMI_PIXCLK
DCMI_VSYNC
EMMC_CK
EMMC_CMD
EMMC_D0
EMMC_D1
EMMC_D2
EMMC_D3
EMMC_RST_N
I2C1_SCL
I2C1_SDA
I2C2_SCL
I2C2_SDA
I2C3_SCL
I2C3_SDA
IMU_CS_N
IMU_INT1
IMU_INT2
IMU_MISO
IMU_MOSI
IMU_SCK
MAG_DRDY
MTQ_INA1
MTQ_INA2
MTQ_INA3
MTQ_INB1
MTQ_INB2
MTQ_INB3
MTQ_ISNS1
MTQ_ISNS2
MTQ_ISNS3
MTQ_nSLEEP
OBC_HB
QSPI_CLK
QSPI_IO0
QSPI_IO1
QSPI_IO2
QSPI_IO3
QSPI_NCS
RTC_CLKOUT
RTC_INT
RW1_RXD
SUND3
UHF_TXD
UMB_RXD
UMB_TXD
S_MISO
UHF_RXD
SUNA1
SUNB1
SUNA3
SUNB3
SDRAM_D2
SDRAM_D3
RW1_TXD
SDRAM_D13
SDRAM_D14
SDRAM_D15
SDRAM_D0
SDRAM_D1
SDRAM_NBL1
SDRAM_D4
SDRAM_D5
SDRAM_D6
SDRAM_D7
SDRAM_D8
SDRAM_D9
SDRAM_D10
SDRAM_D11
SDRAM_D12
SDRAM_A0
SDRAM_A1
SDRAM_A2
SDRAM_A3
SDRAM_A4
SDRAM_A5
SUNA4
SUNB4
SUNC4
SUND4
SDRAM_SDNRAS
SDRAM_A6
SDRAM_A7
SDRAM_A8
SDRAM_A9
S_SCK
S_MOSI
S_EN
RW3_RXD
SDRAM_A10
SDRAM_A11
SDRAM_A12
SDRAM_BA0
SDRAM_BA1
SDRAM_SDCLK
RW2_RXD
S_STATUS
RW2_TXD
SDRAM_SDNCAS
SDRAM_SDCKE0
SDRAM_SDNE0
SDRAM_SDNWE
UHF_EN
UHF_RST_N
S_CS_N
SUNC3
SWCLK
SWDIO
RW3_TXD
SUNC1
SDRAM_NBL0
SUND1
SUNA2
SUNC2
NRST
SUND2
CAM_IF
CAM_IF.SchDoc
DCMI_D0
DCMI_D1
DCMI_D2
DCMI_D3
DCMI_D4
DCMI_D5
DCMI_D6
DCMI_D7
DCMI_D8
DCMI_D9
DCMI_D10
DCMI_D11
DCMI_PIXCLK
DCMI_HSYNC
DCMI_VSYNC
CAM_XCLK
I2C3_SCL
I2C3_SDA
CAM_RST_N
CAM_STBY
CAM_TRIG
CAM_NTC
I2C3_SDA
I2C3_SCL
CAM_RST_N
Repeat(MTQ,1,3)
MTQ_CH.SchDoc
Repeat(COIL_A)
Repeat(COIL_B)
Repeat(ISNS)
Repeat(IN1)
Repeat(IN2)
nSLEEP
Repeat(PANEL,1,4)
PANEL_CH.SchDoc
Repeat(COIL_B)
Repeat(COIL_A)
Repeat(SUN_A)
Repeat(SUN_B)
Repeat(SUN_C)
Repeat(SUN_D)
BUS_IF
BUS_IF.SchDoc
RW3_RXD
RW3_TXD
I2C1_SCL
I2C1_SDA
OBC_HB
NRST
UHF_TXD
UHF_RXD
UHF_EN
UHF_RST_N
S_SCK
S_MOSI
S_MISO
S_CS
S_EN
S_STATUS
SWDIO
SWCLK
NRST
UMB_TXD
UMB_RXD
RW1_TXD
RW1_RXD
RW2_TXD
RW2_RXD
I2C1_SCL
I2C1_SCL
UHF_EN
S_EN
OBC_HB
I2C1_SCL
I2C1_SDA
UMB_TXD
UMB_RXD
NRST
ADCS_SENS
ADCS_SENS.SchDoc
IMU_SCK
IMU_MOSI
IMU_MISO
IMU_CS_N
I2C2_SCL
I2C2_SDA
RTC_INT
RTC_CLKOUT
MAG_DRDY
IMU_INT1
IMU_INT2
PWR_IN
PWR_IN.SchDoc
Title
Size
A4
Number
Revision
Date: 10/26/2026
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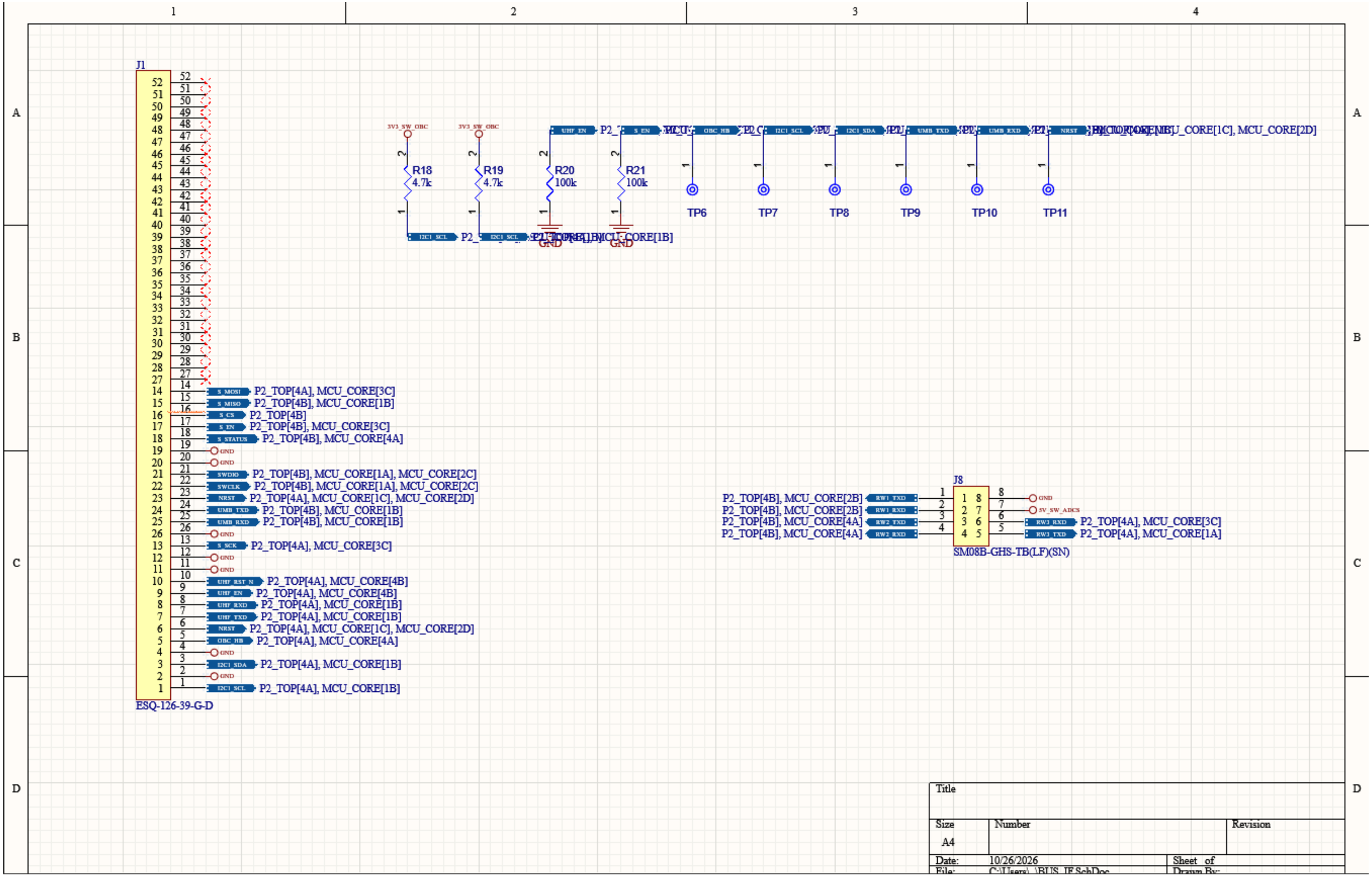

J2
ESQ-126-39-G-D
GND
5V_SW_ADCS
3V3_SW_CAM
3V3_SW_OBC
C12
10µF
C13
10µF
C14
10µF
TP1
TP2
TP3
TP4
TP5
Title
Size
A4
Number
Revision
Date:
10/26/2026
Sheet of
File:
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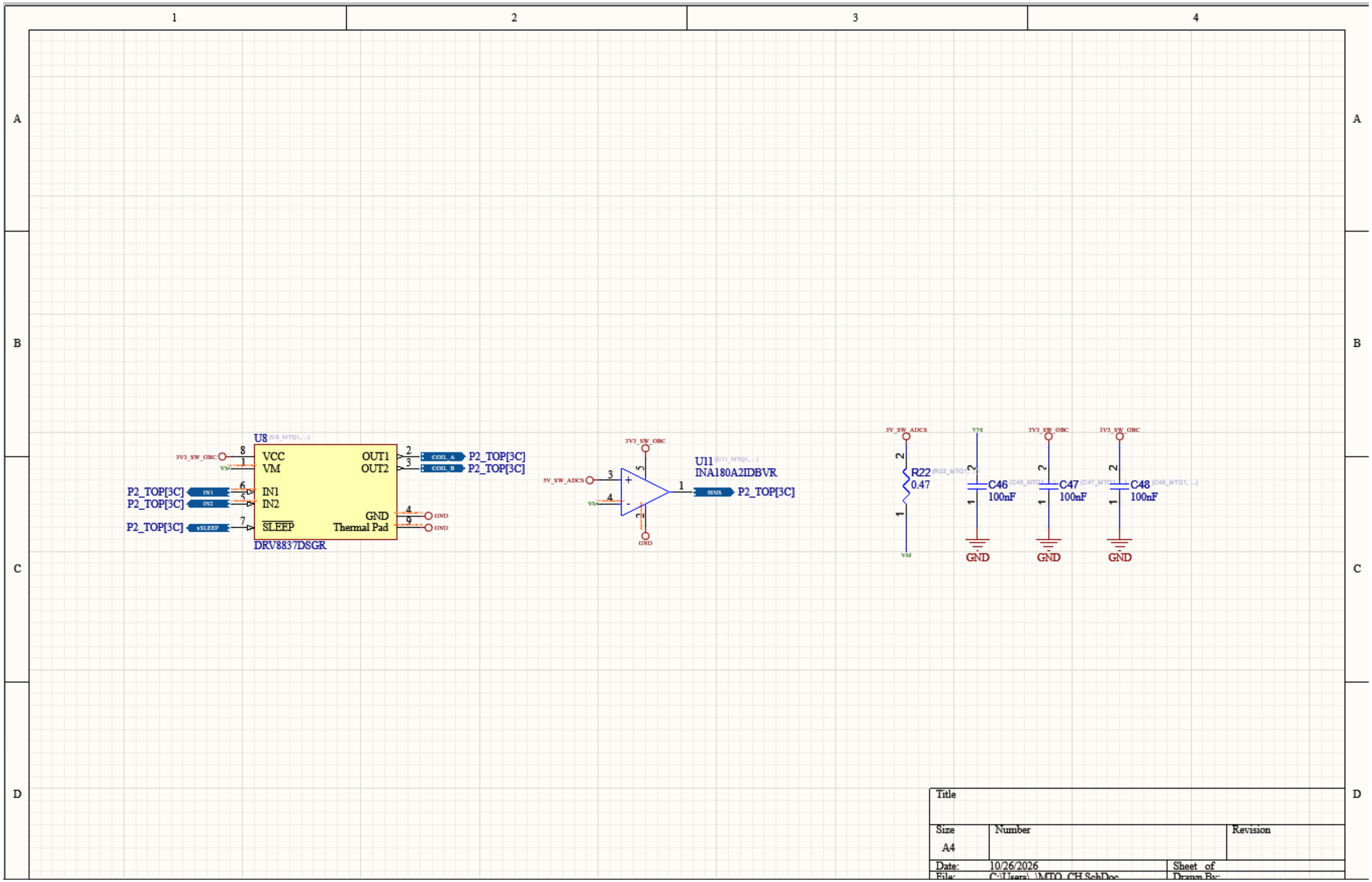

U1B
STM32H743IIT6
U1C
STM32H743IIT6
U1A
STM32H743IIT6
J9
FTSH-105-01-L-DV-K
Y1
ECS-250-8-33-JGN-TR
Title
Size
A4
Number
Revision
Date: 10/26/2026
Sheet of
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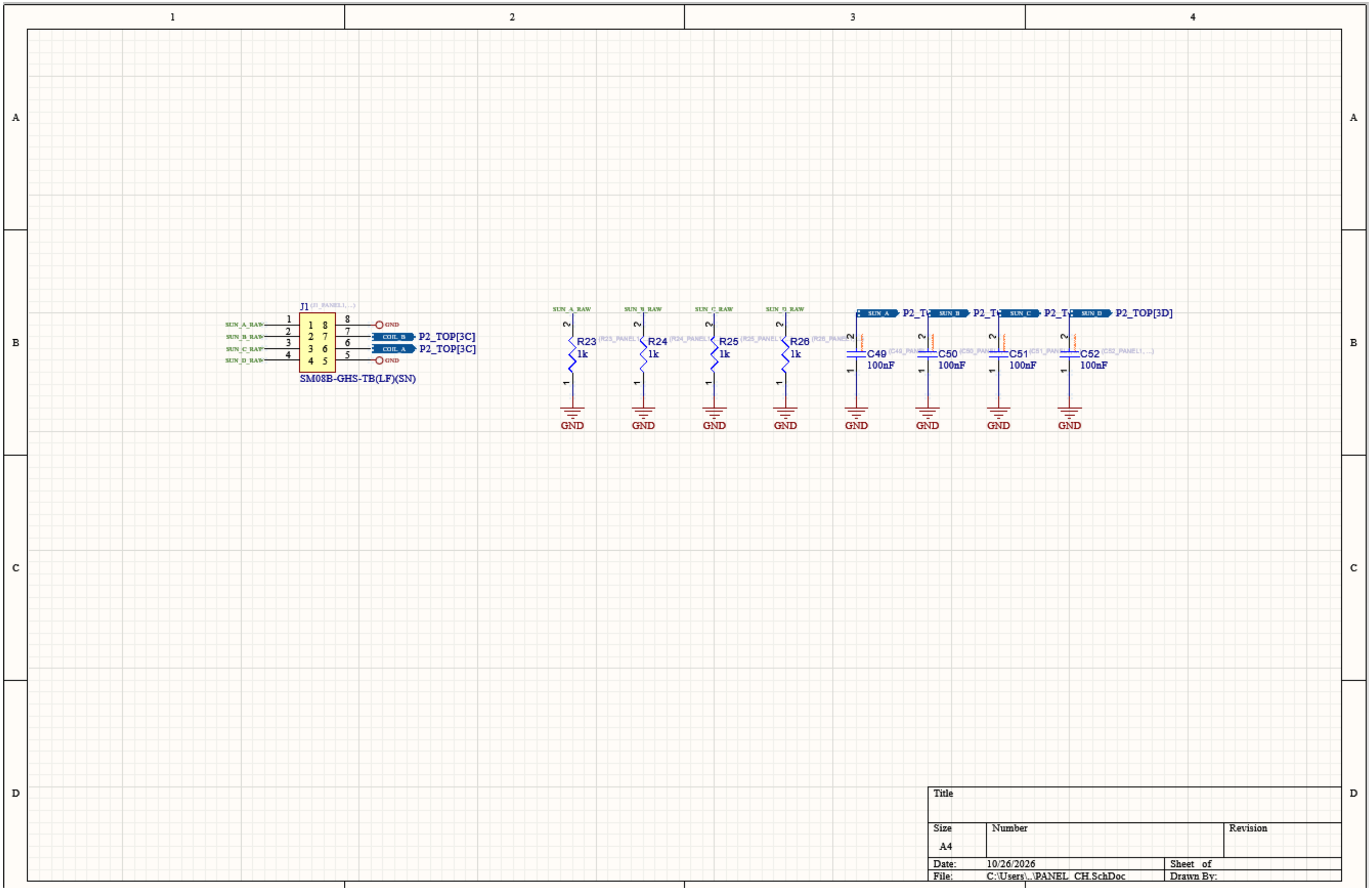

U2
IS42S16160L-6TLI
U4
W25Q128JVSIQ
U3
IS21EF04GP-JCLI
P2_TOP[1D], MCU_CORE[2A]
Title
Size A4
Number
Revision
Date: 10/26/2026
Sheet of
File: C:\Users\...\MEMORY.SchDoc
Drawn By:


1
2
3
4
A
B
C
D
3V3_SW_OBC
3V3_SW_OBC
CAM_RST_N
P2_TOP[3B], MCU_CORE[3C]
R11
4.7k
R12
4.7k
R13
100k
I2C3_SCL
GND
P2_TOP[3A], MCU_CORE[4B]
P2_TOP[3A], MCU_CORE[4A]
P2_TOP[3A], MCU_CORE[3A]
P2_TOP[3A], MCU_CORE[2B]
P2_TOP[3A], MCU_CORE[4B]
P2_TOP[3A], MCU_CORE[3C]
P2_TOP[3A], MCU_CORE[1B]
P2_TOP[3A], MCU_CORE[4B]
P2_TOP[3B], MCU_CORE[1A]
P2_TOP[3B], MCU_CORE[1A]
P2_TOP[3B], MCU_CORE[1B]
P2_TOP[3B], MCU_CORE[1A]
P2_TOP[3B], MCU_CORE[4B]
P2_TOP[3B], MCU_CORE[4B]
P2_TOP[3B], MCU_CORE[3C]
P2_TOP[3B], MCU_CORE[3C]
P2_TOP[3B], MCU_CORE[1B]
P2_TOP[3B], MCU_CORE[4B]
DCMI_D0
DCMI_D1
DCMI_D2
DCMI_D3
DCMI_D4
DCMI_D5
DCMI_D6
DCMI_D7
DCMI_D8
DCMI_D9
DCMI_D10
DCMI_D11
DCMI_PIXCLK
DCMI_HSYNC
DCMI_VSYNC
CAM_XCLK
SCL
SDA
CAM_RST_N
CAM_STBY
CAM_TRIG
3V3_SW_CAM
3V3_SW_CAM
CAM_NTC
MNT 1
MNT 2
J3
FH12-40S-0.5SH(55)
Title
Size
A4
Number
Revision
Date:
10/26/2026
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U5
VDD
VDDIO
INT1
INT2
SCx
SDx
OCS Aux
SDO Aux
SCL
SDA
SDO/SA0
CS
GND
GND
ISM330DHCXTR
P2_TOP[4D], MCU_CORE[2B]
P2_TOP[4D], MCU_CORE[3A]
P2_TOP[4C], MCU_CORE[4A]
P2_TOP[4C], MCU_CORE[2B]
P2_TOP[4C], MCU_CORE[1B]
P2_TOP[4C], MCU_CORE[3C]
U7
SCL
SDA
INT
CLKOUT
EVI
VSS
VBACKUP
VDD
RV-3032-C7
P2_TOP[4C], MCU_CORE[2A]
U6
VDD
VDD IO
C1
CS
SCL/SPC
INT/DRDY/SDO
SDA/SDI/SDO
NC
NC
NC
GND
GND
LIS2MDLTR
P2_TOP[4C], MCU_CORE[4A]
R14
10k
C40
100nF
C41
100nF
C42
100nF
C43
100nF
C44
220nF
C45
100nF
R15
2.2k
R16
2.2k
R17
10k
Title
Size
A4
Number
Revision
Date:
10/26/2026
Sheet of


J1
ESQ-126-39-G-D
P2_TOP[4A], MCU_CORE[3C]
P2_TOP[4B], MCU_CORE[1B]
P2_TOP[4B]
P2_TOP[4B], MCU_CORE[3C]
P2_TOP[4B], MCU_CORE[4A]
P2_TOP[4B], MCU_CORE[1A], MCU_CORE[2C]
P2_TOP[4B], MCU_CORE[1A], MCU_CORE[2C]
P2_TOP[4A], MCU_CORE[1C], MCU_CORE[2D]
P2_TOP[4B], MCU_CORE[1B]
P2_TOP[4B], MCU_CORE[1B]
P2_TOP[4A], MCU_CORE[3C]
P2_TOP[4A], MCU_CORE[4B]
P2_TOP[4A], MCU_CORE[4B]
P2_TOP[4A], MCU_CORE[1B]
P2_TOP[4A], MCU_CORE[1B]
P2_TOP[4A], MCU_CORE[1C], MCU_CORE[2D]
P2_TOP[4A], MCU_CORE[4A]
P2_TOP[4A], MCU_CORE[1B]
P2_TOP[4A], MCU_CORE[1B]
R18 4.7k
R19 4.7k
R20 100k
R21 100k
TP6
TP7
TP8
TP9
TP10
TP11
J8
SM08B-GHS-TB(LF)(SN)
P2_TOP[4B], MCU_CORE[2B]
P2_TOP[4B], MCU_CORE[2B]
P2_TOP[4B], MCU_CORE[4A]
P2_TOP[4B], MCU_CORE[4A]
P2_TOP[4A], MCU_CORE[3C]
P2_TOP[4A], MCU_CORE[1A]
Title
Size A4
Number
Revision
Date: 10/26/2026
Sheet of
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U8
DRV8837DSGR
VCC
VM
IN1
IN2
SLEEP
OUT1
OUT2
GND
Thermal Pad
3V3_SW_OBC
P2_TOP[3C]
COIL_A
COIL_B
GND
U11
INA180A2IDBVR
5V_SW_ADCS
ISNS
R22
0.47
C46
100nF
C47
100nF
C48
100nF
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Size
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2
3
4
A
B
C
D
J1
SUN_A_RAW
SUN_B_RAW
SUN_C_RAW
SUN_D_RAW
GND
COIL_B
COIL_A
P2_TOP[3C]
P2_TOP[3C]
SM08B-GHS-TB(LF)(SN)
R23
R24
R25
R26
1k
C49
C50
C51
C52
100nF
P2_TOP[3D]
Title
Size
A4
Number
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